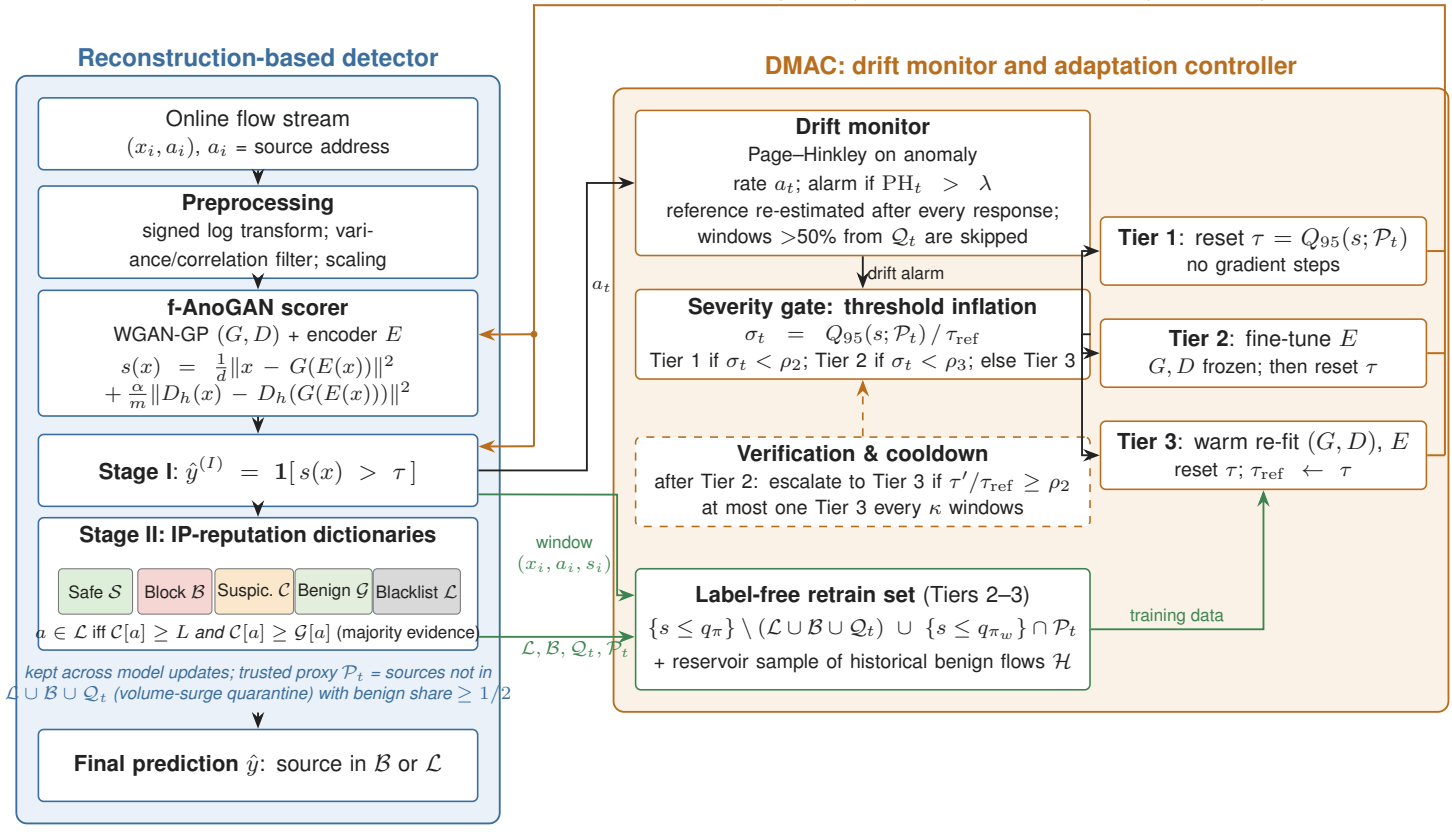

updated model (G, D, E) and threshold τ — reputation state $(\mathcal{S}, \mathcal{B}, \mathcal{C}, \mathcal{G}, \mathcal{L})$ is kept
Reconstruction-based detector
Online flow stream
(x_i, a_i), a_i = source address
Preprocessing
signed log transform; variance/correlation filter; scaling
f-AnoGAN scorer
WGAN-GP (G, D) + encoder E
$s(x) = \frac{1}{d}\|x - G(E(x))\|^2 + \frac{\alpha}{m}\|D_h(x) - D_h(G(E(x)))\|^2$
Stage I: $\hat{y}^{(I)} = \mathbf{1}[\, s(x) > \tau \,]$
Stage II: IP-reputation dictionaries
Safe $\mathcal{S}$
Block $\mathcal{B}$
Suspic. $\mathcal{C}$
Benign $\mathcal{G}$
Blacklist $\mathcal{L}$
$a \in \mathcal{L}$ iff $\mathcal{C}[a] \geq L$ and $\mathcal{C}[a] \geq \mathcal{G}[a]$ (majority evidence)
kept across model updates; trusted proxy $\mathcal{P}_t$ = sources not in $\mathcal{L} \cup \mathcal{B} \cup \mathcal{Q}_t$ (volume-surge quarantine) with benign share $\geq 1/2$
Final prediction $\hat{y}$: source in $\mathcal{B}$ or $\mathcal{L}$
a_t
window (x_i, a_i, s_i)
$\mathcal{L}, \mathcal{B}, \mathcal{Q}_t, \mathcal{P}_t$
DMAC: drift monitor and adaptation controller
Drift monitor
Page–Hinkley on anomaly rate a_t; alarm if $\mathrm{PH}_t > \lambda$
reference re-estimated after every response; windows >50% from $\mathcal{Q}_t$ are skipped
drift alarm
Severity gate: threshold inflation
$\sigma_t = Q_{95}(s; \mathcal{P}_t) / \tau_{\mathrm{ref}}$
Tier 1 if $\sigma_t < \rho_2$; Tier 2 if $\sigma_t < \rho_3$; else Tier 3
Verification & cooldown
after Tier 2: escalate to Tier 3 if $\tau'/\tau_{\mathrm{ref}} \geq \rho_2$
at most one Tier 3 every κ windows
Tier 1: reset $\tau = Q_{95}(s; \mathcal{P}_t)$
no gradient steps
Tier 2: fine-tune E
G, D frozen; then reset τ
Tier 3: warm re-fit (G, D), E
reset τ; $\tau_{\mathrm{ref}} \leftarrow \tau$
Label-free retrain set (Tiers 2–3)
$\{s \leq q_\pi\} \setminus (\mathcal{L} \cup \mathcal{B} \cup \mathcal{Q}_t) \cup \{s \leq q_{\pi_w}\} \cap \mathcal{P}_t$
+ reservoir sample of historical benign flows $\mathcal{H}$
training data